

Grade 8
Unit 5
Lesson 5 Practice

Name _____
Date _____
Period _____

1. Is it possible for a triangle to have sides length of 8 in., 14 in., and 19 in.? Justify your reasoning.

Yes, because $8 + 14 = 22 > 19$, $8 + 19 = 27 > 14$, $14 + 19 = 33 > 8$, so triangle inequality theorem is satisfied.

For questions 2–4, determine if the following sets of side length can form a triangle.

2. 12 cm., 3 cm., 7 cm.

$$12 + 3 = 15 > 7$$

$$12 + 7 = 19 > 3$$

$$7 + 3 = 10 < 12 \rightarrow \text{NO}$$

3. 9 m., 10 m., 14 m.

$$9 + 10 = 19 > 14$$

$$9 + 14 = 23 > 10$$

$$10 + 14 = 24 > 9 \rightarrow \text{YES}$$

4. 5.1 ft., 15 ft., 20 ft.

$$5.1 + 15 = 20.1 > 20$$

$$5.1 + 20 = 25.1 > 15$$

$$20 + 15 = 35 > 5.1 \rightarrow \text{YES}$$

In questions 5–8, two side lengths of a triangle are given. Determine a third side that could be used to form a triangle.

5. 2, 5, 4

$$\{4, 5, 6\}$$

6. 45, 36, 15

$$\{x: 9 < x < 81\}$$

7. 9.4, 6.2, 13.1

$$\{x: 3.2 < x < 15.6\}$$

8. 70, 140, 200

$\{x: 70 < x < 210\}$

9. Ava has four sets of straws to use to form triangles. Which of the sets could not be used?

Set 1: {3 cm, 4 cm, 5 cm}

Set 2: {1 cm, 5 cm, 6 cm}

Set 3: {2 cm, 2 cm, 2 cm}

Set 4: {4 cm, 3 cm, 6 cm}

$5 + 1 = 6$

$3 + 4 = 7 > 5$

$3 + 5 = 8 > 4$

$4 + 5 = 9 > 3$

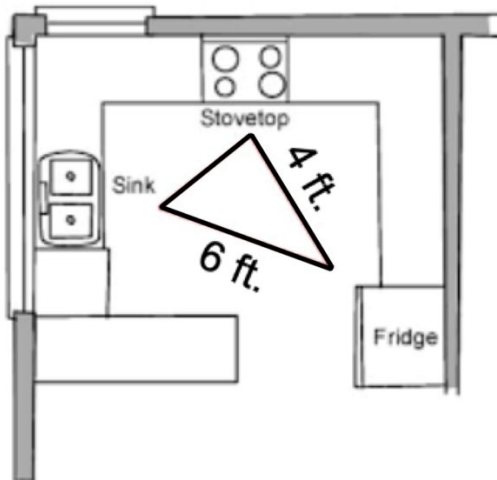
$2 + 2 = 4 > 2$

$4 + 3 = 7 > 6$

$4 + 6 = 10 > 3$

$3 + 6 = 9 > 4$

10. When designing a kitchen, a triangle should be formed between the sink, the oven, and the refrigerator. If the distance from the oven to the refrigerator is 4 feet and the distance from the sink to the refrigerator is 6 feet, determine a possible distance between the sink and the oven.



POSSIBLE ANSWERS:

3, 4, 5, 6, 7, 8, 9

$\{x: 2 < x < 10\}$